

Draft - expert review pending

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Topic 2.1 - Water on Earth, and in Pakistan

A real classroom situation

After the 2022 floods, a teacher in a village near Khairpur gives her class a puzzle. "Three months ago," she says, "this district was under water. Sindh had a third of the country drowned. And this week, the tanker driver says the neighbourhood tank is running low and the price has gone up." A pupil asks the question the whole class is thinking: "How can we have too much water and too little water at the same time?"

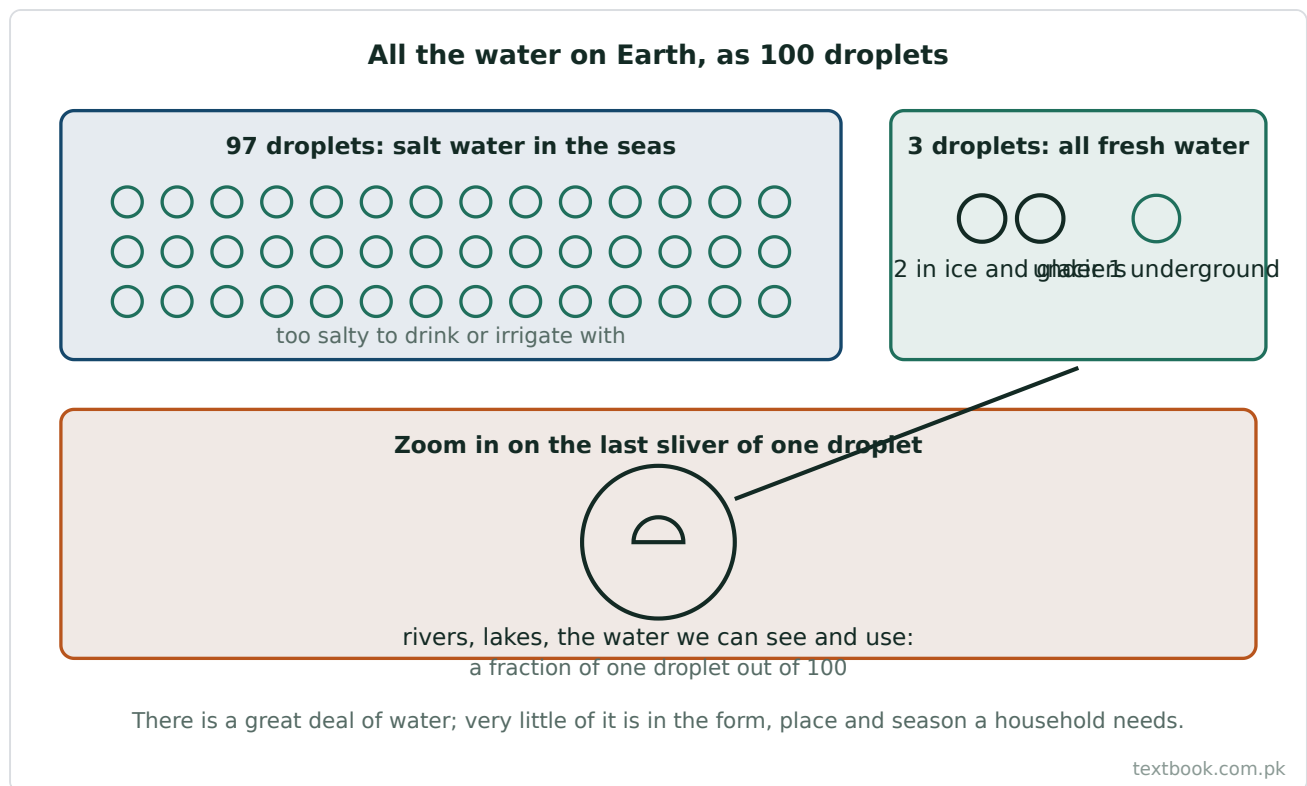
That question is this topic. The answer has two halves: most of Earth's water is not usable where and when people need it, and water security is not about how much water a country receives, but about how much clean water reaches people reliably.

Explanation

Global water distribution, Pakistan's water resources, and water security

Start with the global picture, drawn as 100 droplets. The United States Geological Survey puts the oceans at about 96.5 per cent of all Earth's water, so call it 97 droplets of salt (USGS, 2019). Of the roughly 3 that are fresh, more than two-thirds is locked in ice caps and glaciers, and most of the rest sits underground. The share in rivers and lakes, the water people can see, pump and treat cheaply, is a small fraction of one droplet. There is a great deal of water on

Earth; there is very little of it in the form, the place and the season that a household, a field or a city needs.



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That is why the Khairpur puzzle has an answer. The flood delivered a huge volume of the wrong water, salty in places, contaminated everywhere, arriving in weeks and leaving destruction, while the tanks and wells that supply daily drinking water depend on a steady, clean supply the flood did not provide. Scarcity is a match between need and usable supply, not a count of litres that fell.

Pakistan's water resources are dominated by one system. The Indus and its tributaries, fed by snowmelt and glacial melt and monsoon rain, carry most of the country's surface water; the Indus basin irrigation system is among the largest in the world. Beneath it lie aquifers, recharged slowly, and along the coast the Arabian Sea. Rainfall is uneven: high in the north, thin in Sindh and Balochistan. Researchers analysing the basin describe Pakistan's water resources as already highly stressed, by the ratio of withdrawals to runoff and by per-capita availability, and increasingly so with population growth (Archer et al., 2010).

Pakistan's water resources

Resource	What it supplies	Condition in one line	One challenge
Indus river system	most surface water; the world's largest irrigation network	already highly stressed (withdrawals vs runoff, per-capita availability)	seasonal shifts; upstream changes; growing population
Glaciers and snowmelt	the summer flows that fill the rivers when rain has not fallen	melting and shifting with a warming climate	changed timing of flow; more flood and drought extremes
Groundwater	tube wells and hand pumps where canals do not reach	recharged slowly; can be unsafe, as the Chachro study showed	over-pumping; falling, saltier water tables
Rainfall	monsoon water for dry farming and tank recharge	uneven: high in the north, thin in Sindh and Balochistan	runs off in floods rather than soaking in where needed

Water security = reliable access to acceptable quality and quantity.
Quantity and quality fail separately: the Indus is stressed, Thar's aquifer is unsafe.

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Water scarcity is usually measured per person. Pakistan has crossed from stress toward scarcity as its population has grown, because the same river system now serves more than twice the people it served a generation ago; researchers describe its resources as already highly stressed by per-capita availability (Archer et al., 2010). The World Health Organization reports that over 2 billion people worldwide already live in water-stressed countries (WHO, 2023).

Quality is the second half of security. A well can be abundant and still unsafe. In Taluka Chachro, in the Thar desert, researchers tested thirty-two groundwater samples and found the physicochemical parameters, including total dissolved solids, exceeded WHO drinking-water guideline levels (Samtio et al., 2022). Water security, in the definition this course uses, is reliable access to water of acceptable quality and quantity for health and livelihoods. Quantity and quality fail separately, and a country must manage both.

A worked example. Read a household water bill or a neighbourhood tanker price in Hyderabad, and answer three questions. Where does the water come from (river, canal, groundwater, tanker)? What does the price per litre imply about scarcity? And what happens to it after use, which is Topic 2.2's question. One bill, the whole system.

Common misconception. "Floods mean there is no water scarcity." A flood is a timing and quality failure, not a supply success. The opposite error is also common: "nothing can be done, it is all climate." Both halves are wrong: storage, metering, pricing, treatment and leak repair are all management levers that work whatever the sky does.

Activity: The 100-droplet class

Work in groups of four. Your teacher gives each group 100 small counters (beans, pebbles or bottle caps). Sort them into salt water, ice, groundwater and surface water to match the figure. Then, on paper, split your group's "usable" share among four uses a household makes in a day: drinking and cooking, washing, latrine flushing, and the garden or animals. Argue for your split and defend it to another group. Allow 25 minutes.

Check your understanding

1. Of every 100 units of Earth's water, roughly how many are salt water, and where is most of the fresh water held? (*Remember*)
2. Explain, in two sentences, how a flooded district can still face drinking-water scarcity. (*Understand*)
3. A colleague says Pakistan's water problem is only about rainfall. Use two other parts of the system to show what the statement misses. (*Analyze*)
4. What did the Chachro study find about Thar's groundwater, and which half of water security does it fail, quantity or quality? (*Apply*)

Summary

About 97 per cent of Earth's water is salt; most fresh water is ice and groundwater, and the visible rivers and lakes are a fraction of one per cent. Water security is reliable access to acceptable quality and quantity, so floods and scarcity coexist. Pakistan's resources centre on the Indus system, already highly stressed and increasingly so, plus slowly recharged groundwater that can be abundant yet unsafe, as Thar's Chachro samples showed. Quantity and quality fail separately, and both are managed, not endured.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can draw the 100-droplet picture from memory and label its parts.
- ☐ I can define water security and explain why it has a quality half and a quantity half.
- ☐ I can name Pakistan's four main water resources and one challenge to each.
- ☐ I can explain the flooded-but-scarce puzzle to a pupil in plain words.

Try this at your practicum school

Find out where the school's water comes from: the tap, the tank, the tanker or the hand pump. Ask the clerk or caretaker what it costs per month and whether the supply was cut in the last year. Write half a page naming the source, one quantity fact and one quality fact (colour, smell, taste, or a test the school has done). Do not name the school.

Summative task

Write 150 to 200 words on the following.

Using the 100-droplet picture and the definition of water security, explain to a Class 8 pupil how their district could flood one year and face water shortage the next. Name one thing a household, and one thing a city government, could do to reduce the shortage.

Mini-rubric

Criterion	What a strong response does
Distribution used (Understand)	States the salt/ice/groundwater split correctly rather than vaguely
Security reasoning (Analyze)	Separates quantity from quality and timing, and applies both to the flood case

Criterion	What a strong response does
Levers named (Apply)	One household action and one city action, both realistic in Sindh
Clarity and register	Writes for a Class 8 pupil, plain words; within the word count

Further reading

- U.S. Geological Survey. (2019). *How much water is there on Earth?* (Water Science School): <https://www.usgs.gov/special-topics/water-science-school/science/how-much-water-there-earth>
- Archer, D. R., Forsythe, N., Fowler, H. J., & Shah, S. M. (2010). Sustainability of water resources management in the Indus Basin under changing climatic and socio economic conditions. *Hydrology and Earth System Sciences*, 14(8), 1669-1680. <https://doi.org/10.5194/hess-14-1669-2010>
- Samtio, M. S., Hakro, A. A. D., Mastoi, A. S., & Rajper, R. H. (2022). Groundwater quality assessment for drinking purpose of Taluka Chachro, Thar Desert, Pakistan; using water quality index. *International Journal of Economic and Environmental Geology*, 13(2), 52-57. <https://doi.org/10.46660/ijeeg.Vol13.Iss2.2022.705>
- World Health Organization. (2023). *Drinking-water* (fact sheet): <https://www.who.int/news-room/fact-sheets/detail/drinking-water>
- Holland, W. W. (Ed.). *Oxford Textbook of Public Health*. (guide-recommended book's closest real work, cited by reference only; flagged per D-2026-0001)

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